

Subspace Structure of WeatherGenerator Latent Tokens

Clustering methods and findings

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Abstract. We analyse the 2048-dimensional tokens the WeatherGenerator encoder produces for nine years of ERA5 states (160 M tokens). A K-subspaces clustering, a variant of k-means in which each cluster is a 64-dimensional affine subspace instead of a single centre point, explains 68.5% of token variance with $K=128$ clusters. Plain k-means, run as a baseline on the same data, captures almost none of the structure inside each regime and lands on a different partition. The clusters turn out to be climate regimes. They are geographically coherent, a minority of them carries a strong seasonal cycle, and the El Niño winters stand out in the central Pacific. Several of the largest residual outliers of the fit coincide with documented extreme weather events, so the encoder does capture extremes.

1 Introduction

The WeatherGenerator (WGen) model [1] encodes each 6-hourly ERA5 atmospheric state [2] into 12,288 tokens, one per HEALPix grid cell [3], with 2048 dimensions each. This project analyses the encoder output for all 13,021 available states, covering 2014-01-01 00:00 to 2022-11-30 00:00 UTC. Two questions drove the analysis, in order of importance. First, can we tell from the latents whether the model actually encodes extreme weather events, meaning that real extremes show up as outliers? Second, how does the model handle spatial and temporal structure? Clustering answers both at once. It yields a regime label and a residual per cell and timestep, and the residual is the outlier score.

2 Methodology

Data. 13,021 encoder states of shape $[12,288 \times 2048]$, 1.2 TB in total. All clustering runs use the same fixed random sample of 7,000 states times all cells, 86 M tokens. Evaluation replays the fitted model on the full dataset or on held-out states.

K-subspaces [5, 6]. Cluster j is not a centre μ_j but an affine subspace (μ_j, U_j) with orthonormal basis $U_j \in \mathbb{R}^{2048 \times 64}$. The difference to k-means lies in the assignment step. A token x is assigned by its distance to the whole subspace,

$$r_j(x) = \underbrace{\|x - \mu_j\|^2}_{\text{k-means distance}} - \underbrace{\|U_j^\top (x - \mu_j)\|^2}_{\text{part the subspace explains}}, \quad (1)$$

the centre distance minus what the cluster’s own principal directions account for. A token far from μ_j along directions in which the cluster normally varies still belongs to it. The update step is exact PCA per cluster.

Algorithm (K-subspaces, $K=128$, $d=64$, $D=2048$)

init: $\mu_j \leftarrow K$ random tokens, $U_j \leftarrow 0$

repeat, at most 25 times, until $<0.1\%$ of labels change:

1. assign each token to the nearest subspace:

for every token x and cluster j :

$z \leftarrow U_j^\top (x - \mu_j)$ coords inside the subspace

$r_j(x) \leftarrow \|x - \mu_j\|^2 - \|z\|^2$ Eq. 1

label(x) $\leftarrow \arg \min_j r_j(x)$

2. refit each cluster to its members:

$\mu_j \leftarrow$ mean of its tokens

$U_j \leftarrow$ top- d eigenvectors (PCA) of their covariance

3. split the largest cluster to replace any collapsed one

Setting $d = 0$ turns the same code into plain k-means [4], the first baseline. Greedy k-center [7], which minimises the largest distance to a centre, is the second. A soft-assignment variant (an MPPCA mixture [8]) was fitted afterwards and is used only to quantify per-token uncertainty. One fit takes about two and a half hours on two A40 GPUs. Results were checked on held-out states, across three random restarts, and against an independent run at $K=256$ merged back to 128.

3 Results

3.1 Fit quality and baselines

At $K=128$, $d=64$ the fit splits token variance into 8.4% between cluster centres, 60.1% captured inside the subspaces, and 31.5% residual. Almost all structure is therefore inside the clusters, not between their centres. On held-out dates the residual fraction is also 31.5%, meaning no overfitting.

Table 1 compares the methods. K-means leaves 83.7% of the variance unexplained and finds a different partition (NMI 0.31 against the subspace one). Even when its clusters are given the same 64-dimensional PCA fit after the fact, the residual is still 2270 per token against 1877 to 1892 for the subspace runs. Re-assigning tokens under the fitted subspace model by centre distance alone changes 57% of the labels and adds 20% residual. K-center degenerates. Greedy selection anchors nearly all centres on outliers. One cluster ends up with 98.8% of all tokens (sizes 196 to 84.9M) and the objective lands above the total variance, worse than a single global centroid. Only its minimax radius is stable across seeds (146 to 148).

Table 1: Methods on the identical 86M-token sample, $K=128$. Each row is the best of 3 init seeds (the merge row is a single run). **obj/token** is the mean squared distance from a token to its assigned centre (subspace rows: to its assigned subspace). **% of var** is obj/token divided by the total token variance of 5998, the share of variance left unexplained, lower is better. **spread** is how much worse the worst of the 3 seeds is than the best.

method	obj/token	% of var	spread
k-center (greedy)	9744.2	162%	40.8%
k-means ($d=0$)	5021.1	83.7%	0.05%
K-subspaces ($d=64$)	1887.8	31.5%	0.24%
+ overcluster-merge	1877.0	31.3%	

3.2 Stability and the choice of K

Three restarts from different seeds land within 0.24% in objective and agree pairwise at NMI [10] 0.68 to 0.69 over all 86 M tokens, against below 0.001 for unrelated partitions, and their per-cell dominant-cluster maps agree at 0.72 against a chance level of 0.13. The k-means partitions are less reproducible, their three seeds agreeing only at NMI 0.57 to 0.58 despite a tighter objective spread. Fitting at $K=256$ and merging back to 128 with a Ward-style criterion [9, 11] improves the objective by 0.57%, which is 2.4 times the seed spread, while agreeing with the direct fit only at the seed-to-seed level. The merge cost curve has no knee anywhere between $K=256$ and 32, and 33% of tokens lie within a 10% residual margin of a second cluster. There is no natural number of regimes in this data. K is a resolution choice.

3.3 Extremes appear as outliers

The fit gives a residual per cell and timestep. Per-cell mean residuals span a factor of 12.4, so each cell is standardised against its own history. The top 0.1% of standardised residuals are not noise. 89% of flagged cells have a flagged neighbour (3% for random cells) and 32% persist to the next 6 h step (0.01% random). Four of the largest connected events match documented extremes: the February 2018 sudden-stratospheric-warming Arctic heat anomaly ($z=22.4$) [12], the December 2015 above-freezing North Pole event ($z=19.5$) [13], the August 2016 Arctic cyclone [14], and the March 2022 East-Antarctic heatwave [15, 16]. The single strongest outlier, a late September 2019 event off Tokyo, remains unattributed. The March 2022 event, the largest surface temperature anomaly on record, sits below the global median in raw residual because the Antarctic plateau barely moves in latent space, so only the standardised score recovers such events.

3.4 Clusters have territories

The clustering receives no location or time input, yet the dominant cluster per cell forms contiguous regions that follow tropics, storm tracks, polar caps and continents (Fig. 1). At raw presence level most clusters look global, the most extreme appearing somewhere in 36% of all cells yet holding half its tokens in only 4.4% of the globe. Across the run 77 of 128 clusters reach occupancy above 0.9 somewhere while 12 never pass 0.5. So clusters have clear territories that only a threshold reveals.

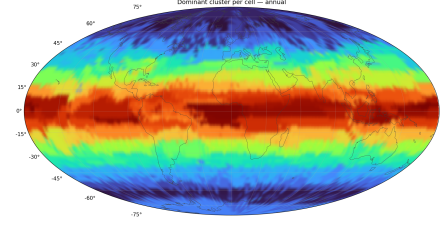


Figure 1: Dominant cluster per cell across the whole 2014 to 2022 record ($K=128$, similar clusters share hues), computed without any location input.

A few clusters encode something specific on their own. Cluster 27 owns the ring around the North Pole between 75 and 86°N, the zone of retreating sea ice, at occupancy 1.0 in its core cells. Its residual is nearly as large as the 6-hour change itself (ratio 0.92, third highest of 128), so the encoder gives this region its own regime but models it comparatively poorly. Fig. 3 shows three territories spanning the range.

3.5 Seasonality and El Niño

Seasonality is real but concentrated in a minority of clusters. The median seasonality score (std/mean of the twelve monthly enrichments) is 0.17, and only 35 of 128 clusters exceed 0.5, mostly winter or summer regimes. Whether that concentration is a property of the atmosphere or partly of the method is open. On the map, 42.8% of cells change their dominant cluster between DJF and JJA (Fig. 2), while the diurnal counterpart is a null (2.6% between 00Z and 12Z). The leading principal component of the per-date regime mix is an almost pure seasonal harmonic ($R^2=0.992$). Interannual events are visible on top of it. In the Niño 3.4 box [17], three of the four El Niño winters of the period (2015/16, 2018/19, 2019/20, all but the weak 2014/15 event) have a mean anomaly score of 0.55 to 0.60 while four of the five remaining winters sit at 0.12 to 0.18. The exception is the 2021/22 La Niña winter at 0.52, so a high score can mark either ENSO phase. During the record 2015/16 event the cluster that usually dominates the box drops to its nine-year occupancy minimum.

4 Discussion

Both guiding questions have positive answers. Documented extremes (SSW 2018, the above-freezing North Pole 2015, Antarctic heatwave 2022) are recoverable from the latents as spatially and temporally coherent outlier events after per-cell standardisation, which is needed because latent residual size is not calibrated to physical anomaly size. Spatially the latents are organised into regimes with real territories, visible only under a mass threshold because almost every cluster appears somewhere on the globe. Temporally the seasonal cycle dominates and is carried by a minority of clearly seasonal clusters, with El Niño winters visible on top of it. Occasionally a single cluster isolates a specific physical region, such as the Arctic sea-ice ring. Finally, the latent space is a continuum with no natural K . Its structure lies in each cluster’s directions of variation rather than in where its centre sits.

Appendix

Both figures come from the same run as Fig. 1, on the same colours. Regenerate with `python3 src/analysis/report_figures.py`, which prints every number the captions quote.

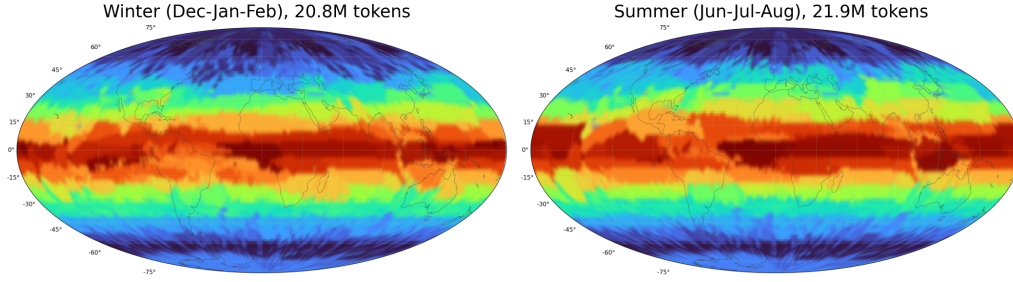


Figure 2: Dominant cluster per cell in winter and in summer (Section 3.5). A region that keeps its colour keeps its regime. 42.8% of cells carry a different dominant cluster in JJA than in DJF, the tropical and northern mid-latitude bands migrating into the summer hemisphere while the hemispheres move in opposite phase. The diurnal version is a null, 2.6% of cells differing between 00Z and 12Z. December is grouped with the following January and February.

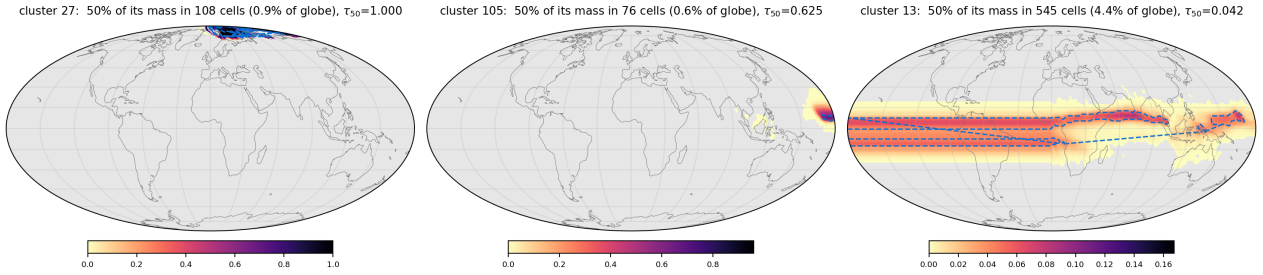


Figure 3: Occupancy of three clusters (Section 3.4), the fraction of the 13,021 time steps at which a cell carries that label. Grey marks cells the cluster never visits and the dashed outline is the smallest set of cells holding half the cluster's tokens. Each panel has its own colour scale because peak occupancy is bimodal, 77 of 128 clusters peaking above 0.9 while 12 never reach 0.5, so one shared scale would hide whichever group it missed. Cluster 27 is a ring around the North Pole that it occupies at every time step, cluster 105 a compact equatorial Pacific regime at the date line and the most common label inside the Niño 3.4 box, cluster 13 the run's most itinerant, somewhere in 36% of all cells while half its mass sits in 4.4% of the globe.

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